

## Questions

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1. A laboratory mixing vessel initially contains 4 litres of solution in which 4 mg of dye is dissolved.

A wash solution enters the vessel at a constant rate of 300 ml per minute.

At the same time, well-mixed liquid leaves the vessel at a constant rate of 200 ml per minute.

The concentration of dye in the liquid entering the vessel, at time  $t$  minutes after pumping begins, is modelled by  $2e^{-0.1t}$  mg per litre.

Let  $m$  be the amount of dye, in milligrams, in the vessel at time  $t$  minutes.

- (a) Show that the volume of liquid in the vessel is  $(4 + 0.1t)$  litres, and hence show that  $m$  satisfies

$$\frac{dm}{dt} = 0.6e^{-0.1t} - \frac{2m}{40 + t} \quad [4]$$

- (b) Solve this differential equation, given that  $m = 4$  when  $t = 0$ , to find an expression for  $m$  in terms of  $t$  [8]

- (c) After 20 minutes, the concentration of dye in the vessel was measured as 0.82 mg per litre. Assess the model in light of this information, giving a reason for your answer [2]

### Solution

- (a) The liquid enters at 300 ml/min and leaves at 200 ml/min, so the net increase in volume is

$$300 - 200 = 100 \text{ ml/min} = 0.1 \text{ L/min}$$

Hence, starting from 4 L, the volume after  $t$  minutes is

$$V(t) = 4 + 0.1t$$

Now form a rate equation for the dye.

The dye enters with concentration  $2e^{-0.1t}$  mg/L and flow rate 0.3 L/min, so the rate of dye entering is

$$0.3 \times 2e^{-0.1t} = 0.6e^{-0.1t} \text{ mg/min}$$

The concentration of dye in the vessel at time  $t$  is

$$\frac{m}{4 + 0.1t} \text{ mg/L}$$

Since liquid leaves at 0.2 L/min, the rate of dye leaving is

$$0.2 \left( \frac{m}{4 + 0.1t} \right)$$

Simplifying,

$$0.2 \left( \frac{m}{4 + 0.1t} \right) = \frac{0.2m}{4 + 0.1t} = \frac{2m}{40 + t}$$

Therefore,

$$\begin{aligned} \frac{dm}{dt} &= \text{rate in} - \text{rate out} \\ &= 0.6e^{-0.1t} - \frac{2m}{40 + t} \end{aligned}$$

So

$$V(t) = 4 + 0.1t \text{ litres,} \quad \frac{dm}{dt} = 0.6e^{-0.1t} - \frac{2m}{40 + t}$$

(b) We solve

$$\frac{dm}{dt} + \frac{2m}{40+t} = 0.6e^{-0.1t}$$

This is a linear differential equation.

The integrating factor is

$$\begin{aligned}\mu(t) &= \exp\left(\int \frac{2}{40+t} dt\right) \\ &= \exp(2 \ln(40+t)) \\ &= (40+t)^2\end{aligned}$$

Multiplying the differential equation by  $(40+t)^2$ ,

$$(40+t)^2 \frac{dm}{dt} + (40+t)^2 \frac{2m}{40+t} = 0.6(40+t)^2 e^{-0.1t}$$

So

$$\frac{d}{dt} ((40+t)^2 m) = 0.6(40+t)^2 e^{-0.1t}$$

Integrating both sides,

$$(40+t)^2 m = \int 0.6(40+t)^2 e^{-0.1t} dt$$

Expand  $(40+t)^2$ :

$$(40+t)^2 = t^2 + 80t + 1600$$

So

$$(40+t)^2 m = \int 0.6(t^2 + 80t + 1600)e^{-0.1t} dt$$

Let

$$I = \int 0.6(t^2 + 80t + 1600)e^{-0.1t} dt$$

and use integration by parts with

$$u = t^2 + 80t + 1600, \quad \frac{dv}{dt} = 0.6e^{-0.1t}$$

Then

$$\frac{du}{dt} = 2t + 80, \quad v = \int 0.6e^{-0.1t} dt = -6e^{-0.1t}$$

So

$$\begin{aligned}I &= uv - \int v \frac{du}{dt} dt \\ &= -6(t^2 + 80t + 1600)e^{-0.1t} + 6 \int (2t + 80)e^{-0.1t} dt \\ &= -6(t^2 + 80t + 1600)e^{-0.1t} + 12 \int (t + 40)e^{-0.1t} dt\end{aligned}$$

Now let

$$J = \int (t + 40)e^{-0.1t} dt$$

Again integrating by parts, take

$$u = t + 40, \quad \frac{dv}{dt} = e^{-0.1t}$$

Then

$$\frac{du}{dt} = 1, \quad v = \int e^{-0.1t} dt = -10e^{-0.1t}$$

Hence

$$\begin{aligned}
 J &= uv - \int v \frac{du}{dt} dt \\
 &= -10(t+40)e^{-0.1t} + 10 \int e^{-0.1t} dt \\
 &= -10(t+40)e^{-0.1t} - 100e^{-0.1t} + K \\
 &= -10(t+50)e^{-0.1t} + K
 \end{aligned}$$

Substituting this into the expression for  $I$ ,

$$\begin{aligned}
 I &= -6(t^2 + 80t + 1600)e^{-0.1t} + 12(-10(t+50)e^{-0.1t}) + K \\
 &= -6e^{-0.1t}(t^2 + 80t + 1600 + 20t + 1000) + K \\
 &= -6e^{-0.1t}(t^2 + 100t + 2600) + K
 \end{aligned}$$

Therefore,

$$\int 0.6(t^2 + 80t + 1600)e^{-0.1t} dt = -6e^{-0.1t}(t^2 + 100t + 2600) + K$$

where  $K$  is a constant.

So

$$(40+t)^2 m = -6e^{-0.1t}(t^2 + 100t + 2600) + K$$

Now use the initial condition  $m = 4$  when  $t = 0$ :

$$\begin{aligned}
 (40)^2(4) &= -6e^0(0^2 + 100(0) + 2600) + K \\
 6400 &= -6(2600) + K \\
 6400 &= -15600 + K
 \end{aligned}$$

Thus

$$K = 22000$$

Hence

$$(40+t)^2 m = 22000 - 6e^{-0.1t}(t^2 + 100t + 2600)$$

and so

$$m = \frac{22000 - 6e^{-0.1t}(t^2 + 100t + 2600)}{(40+t)^2}$$

Therefore, the amount of dye is

$$m(t) = \frac{22000 - 6e^{-0.1t}(t^2 + 100t + 2600)}{(40+t)^2} \text{ mg}$$

(c) After 20 minutes, the model gives volume

$$V(20) = 4 + 0.1(20) = 6 \text{ L}$$

Using the expression for  $m$ ,

$$\begin{aligned}
 m(20) &= \frac{22000 - 6e^{-2}(20^2 + 100(20) + 2600)}{(40+20)^2} \\
 &= \frac{22000 - 6e^{-2}(400 + 2000 + 2600)}{60^2} \\
 &= \frac{22000 - 30000e^{-2}}{3600} \\
 &= \frac{55}{9} - \frac{25}{3}e^{-2}
 \end{aligned}$$

So the predicted concentration is

$$\begin{aligned}\frac{m(20)}{V(20)} &= \frac{1}{6} \left( \frac{55}{9} - \frac{25}{3} e^{-2} \right) \\ &= \frac{55}{54} - \frac{25}{18} e^{-2} \\ &\approx 0.831 \text{ mg/L}\end{aligned}$$

The measured concentration was 0.82 mg/L, which is very close to 0.831 mg/L.

So the model appears reasonable, since its prediction is consistent with the observed value.

2. A particle  $B$ , of mass 2 kg, moves in a straight line and has velocity  $v \text{ m s}^{-1}$ .

At time  $t$  seconds, where  $t \geq 0$ , a variable force of  $\frac{12t - 4v}{1 + t}$  N acts on  $B$ .

There are no other forces acting on  $B$ . Initially, when  $t = 0$ ,  $B$  has velocity  $-5 \text{ m s}^{-1}$ .

The motion of  $B$  can be modelled by the differential equation

$$\frac{dv}{dt} + P(t)v = Q(t)$$

where  $P(t)$  and  $Q(t)$  are functions of  $t$ .

- (a) Find the functions  $P(t)$  and  $Q(t)$  [2]

- (b) Using an integrating factor, determine the first time at which  $B$  is stationary according to the model [8]

### Solution

- (a) Since the mass of  $B$  is 2 kg and the only force acting is

$$\frac{12t - 4v}{1 + t}$$

Newton's second law gives

$$2 \frac{dv}{dt} = \frac{12t - 4v}{1 + t}$$

Rearranging into the form

$$\frac{dv}{dt} + P(t)v = Q(t)$$

gives

$$\begin{aligned} \frac{dv}{dt} &= \frac{12t - 4v}{2(1 + t)} \\ &= \frac{6t - 2v}{1 + t} \\ \frac{dv}{dt} + \frac{2}{1 + t}v &= \frac{6t}{1 + t} \end{aligned}$$

Hence

$$P(t) = \frac{2}{1 + t} \quad \text{and} \quad Q(t) = \frac{6t}{1 + t}$$

- (b) From part (a), the differential equation is

$$\frac{dv}{dt} + \frac{2}{1 + t}v = \frac{6t}{1 + t}$$

The integrating factor is

$$\begin{aligned} \mu(t) &= e^{\int \frac{2}{1+t} dt} \\ &= e^{2 \ln(1+t)} \\ &= (1 + t)^2 \end{aligned}$$

since  $t \geq 0$ , so  $1 + t > 0$ .

Multiplying the differential equation by  $(1 + t)^2$  gives

$$\begin{aligned} (1 + t)^2 \frac{dv}{dt} + \frac{2(1 + t)^2}{1 + t}v &= \frac{6t(1 + t)^2}{1 + t} \\ (1 + t)^2 \frac{dv}{dt} + 2(1 + t)v &= 6t(1 + t) \end{aligned}$$

The left-hand side is now the derivative of  $(1+t)^2v$ , so

$$\frac{d}{dt}((1+t)^2v) = 6t(1+t)$$

Integrating with respect to  $t$ ,

$$\begin{aligned}(1+t)^2v &= \int 6t(1+t) \, dt \\ &= \int (6t + 6t^2) \, dt \\ &= 3t^2 + 2t^3 + C\end{aligned}$$

Use the initial condition  $v = -5$  when  $t = 0$ :

$$\begin{aligned}(1+0)^2(-5) &= 3(0)^2 + 2(0)^3 + C \\ -5 &= C\end{aligned}$$

So

$$(1+t)^2v = 3t^2 + 2t^3 - 5$$

and therefore

$$v = \frac{2t^3 + 3t^2 - 5}{(1+t)^2}$$

Factorising the numerator,

$$2t^3 + 3t^2 - 5 = (t-1)(2t^2 + 5t + 5)$$

so

$$v = \frac{(t-1)(2t^2 + 5t + 5)}{(1+t)^2}$$

The particle is stationary when  $v = 0$ , so we solve

$$\frac{(t-1)(2t^2 + 5t + 5)}{(1+t)^2} = 0$$

For  $t \geq 0$ , we have  $(1+t)^2 > 0$ , and also

$$2t^2 + 5t + 5 > 0$$

so the only way  $v = 0$  is when

$$t - 1 = 0$$

Hence

$$t = 1$$

Therefore, the first time at which  $B$  is stationary is 1 s.

3. A brief flushing cycle in an industrial pipe is to be modelled. The flow rate of liquid,  $R$  litres per second, at time  $t$  seconds is measured from the instant the valve is opened. The cycle starts when  $t = 0$  and there is initially no flow through the pipe. When  $t = 1$ , the flow rate is measured as  $\pi e^{-1} \sin(1)$  litres per second.

While the cycle is occurring,  $R$  is modelled by the differential equation

$$\frac{dR}{dt} + (1 - \cot t)R = \frac{e^{-t} \sin t}{1 + t^2}$$

- (a) By using an integrating factor show that, according to the model, while the cycle is occurring,

$$R = e^{-t} \sin t \left( \tan^{-1} t + \frac{3\pi}{4} \right) \quad [6]$$

The cycle ends when there is again no flow through the pipe.

- (b) Determine the length of time that the cycle lasts according to the model [2]  
 (c) Determine, according to the model, the rate of increase of the flow rate at the start of the cycle. Give your answer in exact form [3]

### Solution

- (a) For the cycle we consider  $0 < t < \pi$ , so  $\sin t > 0$ . The differential equation is

$$\frac{dR}{dt} + (1 - \cot t)R = \frac{e^{-t} \sin t}{1 + t^2}$$

An integrating factor is

$$\begin{aligned} \mu(t) &= e^{\int (1 - \cot t) dt} \\ &= e^{t - \int \cot t dt} \\ &= e^{t - \ln(\sin t)} \\ &= \frac{e^t}{\sin t} \end{aligned}$$

Multiplying the differential equation by  $\frac{e^t}{\sin t}$  gives

$$\begin{aligned} \frac{e^t}{\sin t} \frac{dR}{dt} + \frac{e^t}{\sin t} (1 - \cot t)R &= \frac{e^t}{\sin t} \cdot \frac{e^{-t} \sin t}{1 + t^2} \\ &= \frac{1}{1 + t^2} \end{aligned}$$

Now

$$\begin{aligned} \frac{d}{dt} \left( \frac{e^t}{\sin t} \right) &= \frac{e^t}{\sin t} - \frac{e^t \cos t}{\sin^2 t} \\ &= \frac{e^t}{\sin t} \left( 1 - \frac{\cos t}{\sin t} \right) \\ &= \frac{e^t}{\sin t} (1 - \cot t) \end{aligned}$$

so the left-hand side is

$$\frac{d}{dt} \left( \frac{e^t}{\sin t} R \right)$$

Hence

$$\frac{d}{dt} \left( \frac{e^t}{\sin t} R \right) = \frac{1}{1+t^2}$$

Integrating:

$$\begin{aligned} \frac{e^t}{\sin t} R &= \int \frac{1}{1+t^2} dt \\ &= \tan^{-1} t + C \end{aligned}$$

Therefore

$$R = e^{-t} \sin t (\tan^{-1} t + C)$$

Use the condition  $R(1) = \pi e^{-1} \sin 1$ :

$$\begin{aligned} \pi e^{-1} \sin 1 &= e^{-1} \sin 1 (\tan^{-1} 1 + C) \\ &= e^{-1} \sin 1 \left( \frac{\pi}{4} + C \right) \end{aligned}$$

Since  $e^{-1} \sin 1 \neq 0$ ,

$$\begin{aligned} \pi &= \frac{\pi}{4} + C \\ C &= \frac{3\pi}{4} \end{aligned}$$

So

$$R = e^{-t} \sin t \left( \tan^{-1} t + \frac{3\pi}{4} \right)$$

Thus, according to the model,

$$R = e^{-t} \sin t \left( \tan^{-1} t + \frac{3\pi}{4} \right)$$

(b) The cycle ends when there is again no flow, so we set  $R = 0$ :

$$e^{-t} \sin t \left( \tan^{-1} t + \frac{3\pi}{4} \right) = 0$$

For  $t \geq 0$ , we have  $e^{-t} > 0$  and

$$\tan^{-1} t + \frac{3\pi}{4} > 0$$

so the only way  $R = 0$  is if

$$\sin t = 0$$

Hence

$$t = n\pi \quad (n \in \mathbb{Z})$$

and the first time after  $t = 0$  is

$$t = \pi$$

Therefore the cycle lasts  $\pi$  seconds.

(c) The rate of increase of the flow rate at the start is  $R'(0)$ .

From part (a),

$$R(t) = e^{-t} \sin t \left( \tan^{-1} t + \frac{3\pi}{4} \right)$$

Differentiate:

$$R'(t) = e^{-t} \left[ \cos t \left( \tan^{-1} t + \frac{3\pi}{4} \right) + \frac{\sin t}{1+t^2} - \sin t \left( \tan^{-1} t + \frac{3\pi}{4} \right) \right]$$

Hence

$$R'(0) = 1 \left[ 1 \cdot \frac{3\pi}{4} + 0 - 0 \right] = \frac{3\pi}{4}$$

So the rate of increase of the flow rate at the start is

$$\frac{3\pi}{4} \text{ L s}^{-2}$$

4. A mixing tank initially contains 600 litres of pure water.

Brine flows into the tank at a constant rate of 20 litres per hour. Each litre of brine contains 3 grams of dissolved salt.

The well-stirred mixture is pumped out of the tank at a constant rate of 30 litres per hour.

It is assumed that the salt is instantly and uniformly mixed throughout the liquid.

Given that there are  $x$  grams of salt in the tank after  $t$  hours,

(a) show that, while the tank still contains liquid, the situation can be modelled by

$$\frac{dx}{dt} = 60 - \frac{3x}{60 - t} \quad [4]$$

(b) Hence find the number of grams of salt in the tank after 30 hours [5]

(c) State one limitation of this model and explain briefly how the model could be refined [1]

### Solution

(a) Let the volume of liquid in the tank after  $t$  hours be  $V$  litres.

Since liquid flows in at 20 litres/h and out at 30 litres/h, the net change in volume is

$$V = 600 + 20t - 30t = 600 - 10t = 10(60 - t)$$

This is valid while the tank still contains liquid, so  $0 \leq t < 60$ .

The rate at which salt enters is

$$20 \times 3 = 60 \text{ g/h}$$

The concentration of salt in the tank at time  $t$  is

$$\frac{x}{V} = \frac{x}{600 - 10t} \text{ g/litre}$$

Since the mixture leaves at 30 litres/h, the rate at which salt leaves is

$$30 \left( \frac{x}{600 - 10t} \right) = \frac{30x}{600 - 10t} = \frac{3x}{60 - t} \text{ g/h}$$

Hence,

$$\frac{dx}{dt} = \text{rate in} - \text{rate out}$$

so

$$\frac{dx}{dt} = 60 - \frac{3x}{60 - t}$$

Therefore, while the tank contains liquid,

$$\frac{dx}{dt} = 60 - \frac{3x}{60 - t}$$

(b) We solve

$$\frac{dx}{dt} + \frac{3x}{60 - t} = 60$$

with the initial condition  $x(0) = 0$ , since the tank initially contains pure water.

The integrating factor is

$$\begin{aligned}\mu(t) &= \exp\left(\int \frac{3}{60-t} dt\right) \\ &= \exp(-3\ln(60-t)) \\ &= (60-t)^{-3}\end{aligned}$$

Multiplying the differential equation by  $(60-t)^{-3}$  gives

$$(60-t)^{-3} \frac{dx}{dt} + \frac{3x}{60-t} (60-t)^{-3} = 60(60-t)^{-3}$$

The left-hand side is now the derivative of  $x(60-t)^{-3}$ , so

$$\frac{d}{dt} [x(60-t)^{-3}] = 60(60-t)^{-3}$$

Integrating both sides with respect to  $t$ ,

$$x(60-t)^{-3} = \int 60(60-t)^{-3} dt + C$$

To evaluate the integral, let  $u = 60 - t$ , so  $du = -dt$ :

$$\begin{aligned}\int 60(60-t)^{-3} dt &= -60 \int u^{-3} du \\ &= -60 \left( \frac{u^{-2}}{-2} \right) \\ &= 30u^{-2} \\ &= 30(60-t)^{-2}\end{aligned}$$

Therefore,

$$x(60-t)^{-3} = 30(60-t)^{-2} + C$$

Multiplying through by  $(60-t)^3$ ,

$$x = 30(60-t) + C(60-t)^3$$

Use  $x(0) = 0$ :

$$\begin{aligned}0 &= 30(60) + C(60)^3 \\ 0 &= 1800 + 216000C \\ C &= -\frac{1800}{216000} = -\frac{1}{120}\end{aligned}$$

So

$$x = 30(60-t) - \frac{(60-t)^3}{120}$$

At  $t = 30$ ,

$$\begin{aligned}x(30) &= 30(60-30) - \frac{(60-30)^3}{120} \\ &= 30 \cdot 30 - \frac{30^3}{120} \\ &= 900 - \frac{27000}{120} \\ &= 900 - 225 \\ &= 675\end{aligned}$$

Hence the number of grams of salt in the tank after 30 hours is

$$675 \text{ g}$$

- (c) One limitation is that the model assumes the salt is mixed instantly and uniformly throughout the tank, which is not completely realistic.

A refinement would be to use a model with finite mixing time, for example a compartment model or a model where concentration varies in different parts of the tank.

5. The concentration  $C$ , in suitable units, of a pollutant in a reservoir at time  $t$  hours after a filtration system is switched on is to be modelled by a differential equation. Initially, the concentration is  $A$ , and its long-term value is 0.

- (a) One simple model is to assume that the rate of decrease of concentration is directly proportional to  $C$ .

(i) Formulate a differential equation for this model [1]

(ii) Verify that  $C = Ae^{-kt}$ , where  $k$  is a positive constant, satisfies this differential equation, the initial condition and the long-term condition [3]

An alternative model uses the differential equation

$$\frac{dC}{dt} + \frac{2t}{1+t^2}C = R(t)$$

where  $R(t)$  is a function of  $t$ .

- (b) Find the integrating factor for this differential equation, showing that it can be written in the form

$$1 + t^2 \quad [3]$$

- (c) Suppose that  $R(t) = 0$ .

(i) Show that

$$C = \frac{A}{1+t^2} \quad [4]$$

(ii) Find the time predicted by this model for the concentration to fall to half its initial value. Give your answer correct to the nearest minute [2]

- (d) Now suppose that

$$R(t) = \frac{te^{-t}}{1+t^2}$$

Show that

$$C = \frac{A+1-(1+t)e^{-t}}{1+t^2} \quad [6]$$

It is found that the initial concentration is 5, and the concentration falls to half this value after 63 minutes.

- (e) Determine which of the models proposed in parts (c) and (d) is more consistent with these data [2]

### Solution

- (a) (i) If the rate of decrease of concentration is directly proportional to  $C$ , then

$$-\frac{dC}{dt} \propto C$$

so, for some constant  $k > 0$ ,

$$\frac{dC}{dt} = -kC$$

- (ii) Take

$$C = Ae^{-kt}$$

Then

$$\frac{dC}{dt} = \frac{d}{dt}(Ae^{-kt}) = -kAe^{-kt}$$

Since  $Ae^{-kt} = C$ , this becomes

$$\frac{dC}{dt} = -kC$$

so the differential equation is satisfied.

For the initial condition,

$$C(0) = Ae^{-k \cdot 0} = A$$

so the initial concentration is correct.

For the long-term condition, since  $k > 0$ ,

$$\lim_{t \rightarrow \infty} C = \lim_{t \rightarrow \infty} Ae^{-kt} = A \lim_{t \rightarrow \infty} e^{-kt} = 0$$

Hence  $C = Ae^{-kt}$  satisfies the differential equation, the initial condition and the long-term condition.

(b) For

$$\frac{dC}{dt} + \frac{2t}{1+t^2}C = R(t)$$

the integrating factor is

$$\mu(t) = e^{\int \frac{2t}{1+t^2} dt}$$

To evaluate the integral, let

$$u = 1 + t^2 \quad \Rightarrow \quad du = 2t dt$$

Then

$$\begin{aligned} \int \frac{2t}{1+t^2} dt &= \int \frac{1}{u} du \\ &= \ln u \\ &= \ln(1+t^2) \end{aligned}$$

since  $1+t^2 > 0$ .

Therefore

$$\mu(t) = e^{\ln(1+t^2)} = 1+t^2$$

Hence an integrating factor is  $1+t^2$ .

(c) (i) If  $R(t) = 0$ , the differential equation becomes

$$\frac{dC}{dt} + \frac{2t}{1+t^2}C = 0$$

Multiply through by the integrating factor  $1+t^2$ :

$$(1+t^2)\frac{dC}{dt} + 2tC = 0$$

The left-hand side is the derivative of  $(1+t^2)C$ , so

$$\frac{d}{dt}((1+t^2)C) = 0$$

Integrating gives

$$(1+t^2)C = K$$

where  $K$  is a constant.

Use the initial condition  $C(0) = A$ :

$$(1+0^2)A = K$$

so  $K = A$ . Therefore

$$C = \frac{A}{1+t^2}$$

(ii) Half the initial concentration is  $\frac{A}{2}$ , so set

$$\frac{A}{1+t^2} = \frac{A}{2}$$

Cancelling  $A$ ,

$$\frac{1}{1+t^2} = \frac{1}{2}$$

so

$$1+t^2 = 2$$

$$t^2 = 1$$

Since  $t \geq 0$ ,

$$t = 1 \text{ hour}$$

Therefore the time is 60 minutes.

(d) If

$$R(t) = \frac{te^{-t}}{1+t^2}$$

then the differential equation is

$$\frac{dC}{dt} + \frac{2t}{1+t^2}C = \frac{te^{-t}}{1+t^2}$$

Multiply through by the integrating factor  $1+t^2$ :

$$(1+t^2)\frac{dC}{dt} + 2tC = te^{-t}$$

Hence

$$\frac{d}{dt}((1+t^2)C) = te^{-t}$$

Integrating,

$$(1+t^2)C = \int te^{-t} dt$$

Evaluate the integral by parts. Let

$$u = t, \quad \frac{dv}{dt} = e^{-t}$$

Then

$$\frac{du}{dt} = 1, \quad v = -e^{-t}$$

So

$$\begin{aligned} \int te^{-t} dt &= uv - \int v \frac{du}{dt} dt \\ &= t(-e^{-t}) - \int (-e^{-t}) dt \\ &= -te^{-t} + \int e^{-t} dt \\ &= -te^{-t} - e^{-t} + K \\ &= -(1+t)e^{-t} + K \end{aligned}$$

Therefore

$$(1+t^2)C = -(1+t)e^{-t} + K$$

Use  $C(0) = A$ :

$$\begin{aligned} (1+0^2)A &= -(1+0)e^0 + K \\ A &= -1 + K \end{aligned}$$

so

$$K = A + 1$$

Hence

$$C = \frac{A + 1 - (1 + t)e^{-t}}{1 + t^2}$$

Also, using  $\lim_{t \rightarrow \infty} (1 + t)e^{-t} = 0$ ,

$$\lim_{t \rightarrow \infty} C = \lim_{t \rightarrow \infty} \frac{A + 1 - (1 + t)e^{-t}}{1 + t^2} = 0$$

so this model also has the correct long-term value.

(e) From part (c), model (c) predicts the concentration falls to half its initial value after 60 minutes.

For model (d), with  $A = 5$ ,

$$C = \frac{6 - (1 + t)e^{-t}}{1 + t^2}$$

Half the initial concentration is 2.5, so solve

$$\frac{6 - (1 + t)e^{-t}}{1 + t^2} = 2.5$$

Using a calculator gives

$$t \approx 1.055 \text{ hours}$$

which is

$$1.055 \times 60 \approx 63.3 \text{ minutes}$$

This is much closer to the observed 63 minutes than the 60 minutes from model (c). Therefore model (d) is more consistent with the data.

6. A processing tank initially contains 15 litres of water in which 6 mg of dye is dissolved.

A dye solution enters the tank at a constant rate of 200 ml per minute. The concentration of dye in this incoming solution, at time  $t$  minutes after the flow begins, is modelled by  $(4 + e^{-0.1t})$  mg per litre.

At the same time, pure water enters through a second pipe at a constant rate of 100 ml per minute. The liquid in the tank is continually mixed and leaves the tank at a constant rate of 300 ml per minute.

Let  $m$  be the amount of dye, in milligrams, in the tank at time  $t$  minutes after the flow begins.

- (a) (i) Write down the volume of liquid in the tank at time  $t$   
 (ii) Hence show that  $m$  satisfies the differential equation

$$\frac{dm}{dt} = 0.8 + 0.2e^{-0.1t} - \frac{m}{50} \quad [4]$$

- (b) By solving the differential equation, find an expression for  $m$  in terms of  $t$  [8]

After 30 minutes, the concentration of dye in the tank was measured as 1.52 mg per litre.

- (c) Assess the model in light of this information, giving a reason for your answer [2]

### Solution

- (a) (i) Convert the flow rates to litres per minute:

$$200 \text{ ml min}^{-1} = 0.2 \text{ L min}^{-1}, \quad 100 \text{ ml min}^{-1} = 0.1 \text{ L min}^{-1}, \quad 300 \text{ ml min}^{-1} = 0.3 \text{ L min}^{-1}$$

So the net rate of change of volume is

$$0.2 + 0.1 - 0.3 = 0 \text{ L min}^{-1}$$

Hence the volume stays constant at its initial value of 15 litres:

$$V(t) = 15$$

So the volume of liquid in the tank at time  $t$  is 15 litres.

- (ii) The rate at which dye enters through the first pipe is

$$0.2(4 + e^{-0.1t}) \text{ mg min}^{-1}$$

so

$$\text{rate in} = 0.8 + 0.2e^{-0.1t}$$

The second pipe contains pure water, so it contributes no dye.

Since the tank is well mixed and has volume 15 L, the concentration of dye in the tank is

$$\frac{m}{15} \text{ mg L}^{-1}$$

Therefore the rate at which dye leaves the tank is

$$0.3 \times \frac{m}{15} = \frac{0.3m}{15} = \frac{m}{50} \text{ mg min}^{-1}$$

Hence

$$\frac{dm}{dt} = \text{rate in} - \text{rate out}$$

so

$$\frac{dm}{dt} = 0.8 + 0.2e^{-0.1t} - \frac{m}{50}$$

as required.

(b) Rearranging the differential equation into linear form gives

$$\frac{dm}{dt} + \frac{1}{50}m = 0.8 + 0.2e^{-0.1t}$$

The integrating factor is

$$\exp\left(\int \frac{1}{50} dt\right) = e^{t/50}$$

Multiply the differential equation by  $e^{t/50}$ :

$$\begin{aligned} e^{t/50} \frac{dm}{dt} + \frac{1}{50} e^{t/50} m &= 0.8e^{t/50} + 0.2e^{-0.1t} e^{t/50} \\ &= 0.8e^{t/50} + 0.2e^{-0.08t} \end{aligned}$$

The left-hand side is

$$\frac{d}{dt}(me^{t/50})$$

so

$$\frac{d}{dt}(me^{t/50}) = 0.8e^{t/50} + 0.2e^{-0.08t}$$

Integrating both sides with respect to  $t$ :

$$\begin{aligned} me^{t/50} &= \int 0.8e^{t/50} dt + \int 0.2e^{-0.08t} dt + C \\ &= 0.8(50)e^{t/50} + 0.2\left(\frac{1}{-0.08}\right)e^{-0.08t} + C \\ &= 40e^{t/50} - 2.5e^{-0.08t} + C \end{aligned}$$

Divide through by  $e^{t/50}$ :

$$\begin{aligned} m &= 40 - 2.5e^{-0.08t}e^{-t/50} + Ce^{-t/50} \\ &= 40 - 2.5e^{-0.1t} + Ce^{-0.02t} \end{aligned}$$

Initially there are 6 mg of dye in the tank, so  $m(0) = 6$ . Substituting  $t = 0$  gives

$$\begin{aligned} 6 &= 40 - 2.5 + C \\ 6 &= 37.5 + C \\ C &= -31.5 \end{aligned}$$

Therefore

$$m = 40 - 2.5e^{-0.1t} - 31.5e^{-0.02t}$$

(c) The model predicts that at  $t = 30$  minutes,

$$m(30) = 40 - 2.5e^{-3} - 31.5e^{-0.6}$$

so the concentration is

$$\frac{m(30)}{15} = \frac{40 - 2.5e^{-3} - 31.5e^{-0.6}}{15} \approx 1.51 \text{ mg L}^{-1}$$

The measured concentration was  $1.52 \text{ mg L}^{-1}$ , which is very close to the model prediction. Therefore the model appears to be reasonable.

7. A particle  $B$ , of mass 2 kg, moves in a straight line and has velocity  $v \text{ m s}^{-1}$ .

At time  $t$  seconds, where  $t \geq 0$ , a variable force of  $-2t(v + 3) \text{ N}$  acts on  $B$ . There are no other forces acting on  $B$ . Initially, when  $t = 0$ ,  $B$  has velocity  $5 \text{ m s}^{-1}$ .

The motion of  $B$  can be modelled by the differential equation

$$\frac{dv}{dt} + P(t)v = Q(t)$$

where  $P(t)$  and  $Q(t)$  are functions of  $t$ .

(a) Find the functions  $P(t)$  and  $Q(t)$  [2]

(b) Using an integrating factor, determine the first time at which  $B$  is stationary according to the model [8]

### Solution

(a) By Newton's second law, resultant force = mass  $\times$  acceleration, so

$$2 \frac{dv}{dt} = -2t(v + 3)$$

Expanding and dividing by 2 gives

$$\frac{dv}{dt} = -tv - 3t$$

Hence

$$\frac{dv}{dt} + tv = -3t$$

Therefore,  $P(t) = t$  and  $Q(t) = -3t$ .

(b) From part (a), the differential equation is

$$\frac{dv}{dt} + tv = -3t$$

The integrating factor is

$$e^{\int t \, dt} = e^{t^2/2}$$

Multiplying the differential equation by  $e^{t^2/2}$  gives

$$e^{t^2/2} \frac{dv}{dt} + tve^{t^2/2} = -3te^{t^2/2}$$

The left-hand side is

$$\frac{d}{dt} (ve^{t^2/2})$$

so

$$\frac{d}{dt} (ve^{t^2/2}) = -3te^{t^2/2}$$

Integrating with respect to  $t$ ,

$$\begin{aligned} ve^{t^2/2} &= \int -3te^{t^2/2} \, dt \\ &= -3e^{t^2/2} + C \end{aligned}$$

Therefore

$$v = -3 + Ce^{-t^2/2}$$

Using the initial condition  $v = 5$  when  $t = 0$ ,

$$5 = -3 + Ce^0$$

$$5 = -3 + C$$

$$C = 8$$

Hence

$$v = -3 + 8e^{-t^2/2}$$

For  $B$  to be stationary,  $v = 0$ . So

$$0 = -3 + 8e^{-t^2/2}$$

$$8e^{-t^2/2} = 3$$

$$e^{-t^2/2} = \frac{3}{8}$$

Taking natural logarithms,

$$-\frac{t^2}{2} = \ln\left(\frac{3}{8}\right)$$

$$\frac{t^2}{2} = \ln\left(\frac{8}{3}\right)$$

$$t^2 = 2\ln\left(\frac{8}{3}\right)$$

Since  $t \geq 0$ , the first time is

$$t = \sqrt{2\ln\left(\frac{8}{3}\right)}$$

Numerically,

$$t \approx 1.40 \text{ s}$$

Therefore, the first time at which  $B$  is stationary is  $\sqrt{2\ln\left(\frac{8}{3}\right)}$  s, approximately 1.40 s.

8. A small rescue boat moves in a straight line on calm water. It starts from rest, is driven by its engine for 5 seconds and then, for the next 5 seconds, the engine is cut and a drogue is deployed. The boat comes to rest at the end of the 10 seconds. The total mass of the boat and crew is  $m$  kg, and at time  $t$  seconds, for  $0 \leq t \leq 10$ , the boat's velocity is  $v$  m s<sup>-1</sup>.

A resistance to motion, modelled by a force of magnitude  $0.2mv$  N, acts on the boat throughout the 10 seconds.

- (a) Explain why modelling the resistance to motion in this way is likely to be more realistic than assuming this force is constant [1]
- (b) During the drogue phase of the motion, for  $5 \leq t \leq 10$ , the drogue exerts an additional constant resistance force of magnitude  $m$  N and the engine provides no driving force. Show that, for  $5 \leq t \leq 10$ ,

$$\frac{dv}{dt} + 0.2v = -1 \quad [1]$$

- (c) (i) Solve the differential equation in part (b) [4]  
 (ii) Hence find the velocity of the boat when  $t = 5$  [1]

- (d) During the powered phase ( $0 \leq t \leq 5$ ), the propeller provides a driving force of magnitude directly proportional to  $t^2$ . Show that, for  $0 \leq t \leq 5$ ,

$$\frac{dv}{dt} + 0.2v = \lambda t^2 \quad [1]$$

where  $\lambda$  is a positive constant

- (e) (i) Show by integration that, for  $0 \leq t \leq 5$ ,

$$v = 5\lambda(t^2 - 10t + 50 - 50e^{-t/5}) \quad [6]$$

- (ii) Hence find  $\lambda$  [2]

- (f) Find the total distance, to the nearest metre, travelled by the boat during the motion [6]

### Solution

- (a) A resistive force depending on  $v$  is more realistic because water resistance increases as speed increases, and when  $v = 0$  this model gives zero resistance. A constant resistive force would not reflect either of these features.

- (b) During  $5 \leq t \leq 10$ , the only horizontal forces are the two resistances:

$$0.2mv \quad \text{and} \quad m$$

both acting opposite to the motion. Taking the direction of motion as positive and using Newton's second law,

$$m \frac{dv}{dt} = -0.2mv - m$$

Dividing by  $m$  gives

$$\frac{dv}{dt} + 0.2v = -1$$

as required.

- (c) (i) For  $5 \leq t \leq 10$ , we solve

$$\frac{dv}{dt} + \frac{1}{5}v = -1$$

The integrating factor is

$$e^{\int \frac{1}{5} dt} = e^{t/5}$$

Multiplying the differential equation by  $e^{t/5}$ ,

$$\begin{aligned} e^{t/5} \frac{dv}{dt} + \frac{1}{5} e^{t/5} v &= -e^{t/5} \\ \frac{d}{dt} (ve^{t/5}) &= -e^{t/5} \end{aligned}$$

Integrating,

$$\begin{aligned} ve^{t/5} &= \int -e^{t/5} dt \\ &= -5e^{t/5} + C \end{aligned}$$

Hence

$$v = -5 + Ce^{-t/5}$$

The boat comes to rest at  $t = 10$ , so  $v = 0$  when  $t = 10$ :

$$\begin{aligned} 0 &= -5 + Ce^{-2} \\ Ce^{-2} &= 5 \\ C &= 5e^2 \end{aligned}$$

Therefore

$$v = 5e^2 e^{-t/5} - 5 = 5(e^{2-t/5} - 1)$$

So the solution is  $v = 5(e^{2-t/5} - 1)$  for  $5 \leq t \leq 10$ .

(ii) Substituting  $t = 5$  into the result from part (i),

$$\begin{aligned} v(5) &= 5(e^{2-1} - 1) \\ &= 5(e - 1) \end{aligned}$$

Hence the velocity when  $t = 5$  is  $5(e - 1) \text{ m s}^{-1}$ .

- (d) Let the driving force during  $0 \leq t \leq 5$  be  $\lambda mt^2$ , where  $\lambda > 0$  is a constant. Then Newton's second law gives

$$m \frac{dv}{dt} = \lambda mt^2 - 0.2mv$$

Dividing by  $m$ ,

$$\frac{dv}{dt} + 0.2v = \lambda t^2$$

as required.

- (e) (i) We solve

$$\frac{dv}{dt} + \frac{1}{5}v = \lambda t^2$$

The integrating factor is again

$$e^{t/5}$$

Multiplying through by  $e^{t/5}$ ,

$$\begin{aligned} e^{t/5} \frac{dv}{dt} + \frac{1}{5} e^{t/5} v &= \lambda t^2 e^{t/5} \\ \frac{d}{dt} (ve^{t/5}) &= \lambda t^2 e^{t/5} \end{aligned}$$

So

$$ve^{t/5} = \lambda \int t^2 e^{t/5} dt + C$$

Let

$$I = \int t^2 e^{t/5} dt$$

Integrating by parts with  $u = t^2$  and  $\frac{dw}{dt} = e^{t/5}$ ,

$$\begin{aligned} I &= 5t^2 e^{t/5} - \int 10te^{t/5} dt \\ &= 5t^2 e^{t/5} - 10 \int te^{t/5} dt \end{aligned}$$

Now let

$$J = \int te^{t/5} dt$$

Again integrating by parts,

$$\begin{aligned} J &= 5te^{t/5} - \int 5e^{t/5} dt \\ &= 5te^{t/5} - 25e^{t/5} \end{aligned}$$

Hence

$$\begin{aligned} I &= 5t^2 e^{t/5} - 10 \left( 5te^{t/5} - 25e^{t/5} \right) \\ &= 5t^2 e^{t/5} - 50te^{t/5} + 250e^{t/5} \\ &= 5e^{t/5} (t^2 - 10t + 50) \end{aligned}$$

Therefore

$$\begin{aligned} ve^{t/5} &= \lambda \left[ 5e^{t/5} (t^2 - 10t + 50) \right] + C \\ v &= 5\lambda (t^2 - 10t + 50) + Ce^{-t/5} \end{aligned}$$

The boat starts from rest, so  $v = 0$  when  $t = 0$ :

$$\begin{aligned} 0 &= 5\lambda(50) + C \\ C &= -250\lambda \end{aligned}$$

Hence

$$\begin{aligned} v &= 5\lambda (t^2 - 10t + 50) - 250\lambda e^{-t/5} \\ &= 5\lambda \left( t^2 - 10t + 50 - 50e^{-t/5} \right) \end{aligned}$$

Thus, for  $0 \leq t \leq 5$ ,

$$v = 5\lambda \left( t^2 - 10t + 50 - 50e^{-t/5} \right)$$

- (ii) At  $t = 5$ , the expressions for  $v$  from parts (c) and (e)(i) must be equal.  
From part (e)(i),

$$\begin{aligned} v(5) &= 5\lambda (25 - 50 + 50 - 50e^{-1}) \\ &= 5\lambda \left( 25 - \frac{50}{e} \right) \end{aligned}$$

From part (c)(ii),

$$v(5) = 5(e - 1)$$

Equating these,

$$\begin{aligned} 5\lambda \left( 25 - \frac{50}{e} \right) &= 5(e - 1) \\ \lambda \left( 25 - \frac{50}{e} \right) &= e - 1 \\ \lambda \left( \frac{25e - 50}{e} \right) &= e - 1 \\ \lambda \left( \frac{25(e - 2)}{e} \right) &= e - 1 \end{aligned}$$

So

$$\lambda = \frac{e(e-1)}{25(e-2)}$$

Therefore

$$\lambda = \frac{e(e-1)}{25(e-2)} \approx 0.260$$

(f) Since the boat moves forwards throughout the motion, the total distance is

$$s = \int_0^{10} v \, dt = \int_0^5 v \, dt + \int_5^{10} v \, dt$$

For the powered phase,

$$\begin{aligned} s_1 &= \int_0^5 5\lambda (t^2 - 10t + 50 - 50e^{-t/5}) \, dt \\ &= 5\lambda \left[ \frac{t^3}{3} - 5t^2 + 50t + 250e^{-t/5} \right]_0^5 \\ &= 5\lambda \left( \frac{125}{3} - 125 + 250 + \frac{250}{e} - 250 \right) \\ &= 5\lambda \left( \frac{125}{3} - 125 + \frac{250}{e} \right) \\ &= 5\lambda \left( -\frac{250}{3} + \frac{250}{e} \right) \\ &= 1250\lambda \left( \frac{3-e}{3e} \right) \end{aligned}$$

Substituting  $\lambda = \frac{e(e-1)}{25(e-2)}$ ,

$$\begin{aligned} s_1 &= 1250 \left( \frac{e(e-1)}{25(e-2)} \right) \left( \frac{3-e}{3e} \right) \\ &= \frac{50(e-1)(3-e)}{3(e-2)} \end{aligned}$$

So

$$s_1 \approx 11.2 \text{ m}$$

For the drogue phase,

$$\begin{aligned} s_2 &= \int_5^{10} 5(e^{2-t/5} - 1) \, dt \\ &= 5 \left[ \int_5^{10} e^{2-t/5} \, dt - \int_5^{10} 1 \, dt \right] \\ &= 5 \left[ -5e^{2-t/5} - t \right]_5^{10} \\ &= 5((-5 - 10) - (-5e - 5)) \\ &= 25(e - 2) \end{aligned}$$

So

$$s_2 \approx 18.0 \text{ m}$$

Therefore the total distance is

$$\begin{aligned} s &= s_1 + s_2 \\ &= \frac{50(e-1)(3-e)}{3(e-2)} + 25(e-2) \\ &\approx 11.2 + 18.0 \\ &\approx 29.2 \end{aligned}$$

Hence the total distance travelled, to the nearest metre, is 29 m.

9. A winch raises a lift cage vertically. The cage starts from rest and comes to rest again 8 seconds later. The combined mass of the cage and its load is  $m$  kg, and at time  $t$  seconds, for  $0 \leq t \leq 8$ , the cage has upward velocity  $v \text{ m s}^{-1}$ .  
A resistance to the motion, modelled by a force of magnitude  $0.25mv$  N, acts on the cage throughout the 8 seconds.

(a) Explain why modelling the resistance to the motion in this way is likely to be more realistic than assuming this force is constant [1]

(b) During the final stage of the motion, for  $4 \leq t \leq 8$ , the winch provides a constant upward tension of magnitude  $0.75mg$  N. Show that, for  $4 \leq t \leq 8$ ,

$$\frac{dv}{dt} + 0.25v = -\frac{1}{4}g \quad [1]$$

(c) (i) Solve the differential equation in part (b) [5]

(ii) Hence find the velocity of the cage when  $t = 4$  [1]

(d) During the first stage of the motion, for  $0 \leq t \leq 4$ , the winch provides an upward tension of magnitude  $m(g + \lambda t)$  N, where  $\lambda$  is a positive constant. Show that, for  $0 \leq t \leq 4$ ,

$$\frac{dv}{dt} + 0.25v = \lambda t \quad [1]$$

(e) (i) Show by integration that, for  $0 \leq t \leq 4$ ,

$$v = 4\lambda(t - 4 + 4e^{-t/4}) \quad [5]$$

(ii) Hence find  $\lambda$  [2]

(f) Find the total distance, to the nearest metre, through which the cage rises during the motion [6]

### Solution

(a) A resistive force usually depends on the speed of the moving object. When the cage moves faster, the resistance is greater, so modelling it as proportional to  $v$  is more realistic than taking it to be constant.

(b) For  $4 \leq t \leq 8$ , taking upwards as positive:

$$\text{tension} = 0.75mg, \quad \text{weight} = mg, \quad \text{resistance} = 0.25mv$$

Using Newton's second law,

$$\begin{aligned} m \frac{dv}{dt} &= 0.75mg - mg - 0.25mv \\ &= -0.25mg - 0.25mv \end{aligned}$$

Dividing by  $m$ ,

$$\frac{dv}{dt} = -0.25g - 0.25v$$

Hence

$$\frac{dv}{dt} + 0.25v = -\frac{1}{4}g$$

(c) (i) For  $4 \leq t \leq 8$ , solve

$$\frac{dv}{dt} + 0.25v = -\frac{1}{4}g$$

The integrating factor is

$$e^{\int 0.25 dt} = e^{t/4}$$

Multiplying the differential equation by  $e^{t/4}$ ,

$$e^{t/4} \frac{dv}{dt} + 0.25e^{t/4}v = -\frac{1}{4}ge^{t/4}$$

The left-hand side is

$$\frac{d}{dt} (ve^{t/4})$$

so

$$\frac{d}{dt} (ve^{t/4}) = -\frac{1}{4}ge^{t/4}$$

Integrating,

$$\begin{aligned} ve^{t/4} &= \int -\frac{1}{4}ge^{t/4} dt \\ &= -ge^{t/4} + C \end{aligned}$$

Hence

$$v = Ce^{-t/4} - g$$

The cage comes to rest at  $t = 8$ , so  $v(8) = 0$ . Therefore

$$0 = Ce^{-8/4} - g$$

$$0 = Ce^{-2} - g$$

$$C = ge^2$$

Substituting for  $C$ ,

$$\begin{aligned} v &= ge^2e^{-t/4} - g \\ &= g(e^{2-t/4} - 1) \end{aligned}$$

So, for  $4 \leq t \leq 8$ ,

$$v = g(e^{2-t/4} - 1)$$

(ii) Using the result from part (i),

$$\begin{aligned} v(4) &= g(e^{2-4/4} - 1) \\ &= g(e^1 - 1) \\ &= g(e - 1) \end{aligned}$$

With  $g = 9.8$ ,

$$v(4) \approx 9.8(e - 1) \approx 16.8 \text{ m s}^{-1}$$

So the velocity when  $t = 4$  is  $g(e - 1) \text{ m s}^{-1} \approx 16.8 \text{ m s}^{-1}$ .

(d) For  $0 \leq t \leq 4$ , the upward tension is  $m(g + \lambda t)$ .

Using Newton's second law with upwards positive,

$$\begin{aligned} m \frac{dv}{dt} &= m(g + \lambda t) - mg - 0.25mv \\ &= m\lambda t - 0.25mv \end{aligned}$$

Dividing by  $m$ ,

$$\frac{dv}{dt} + 0.25v = \lambda t$$

(e) (i) For  $0 \leq t \leq 4$ , solve

$$\frac{dv}{dt} + 0.25v = \lambda t$$

The integrating factor is

$$e^{\int 0.25 dt} = e^{t/4}$$

Multiplying through by  $e^{t/4}$ ,

$$e^{t/4} \frac{dv}{dt} + 0.25e^{t/4}v = \lambda te^{t/4}$$

Hence

$$\frac{d}{dt} (ve^{t/4}) = \lambda te^{t/4}$$

Integrating,

$$ve^{t/4} = \lambda \int te^{t/4} dt + C$$

Evaluate the integral by parts. Let

$$u = t, \quad \frac{dw}{dt} = e^{t/4}$$

so that

$$\frac{du}{dt} = 1, \quad w = 4e^{t/4}$$

Then

$$\begin{aligned} \int te^{t/4} dt &= 4te^{t/4} - \int 4e^{t/4} dt \\ &= 4te^{t/4} - 16e^{t/4} \end{aligned}$$

Therefore

$$ve^{t/4} = \lambda (4te^{t/4} - 16e^{t/4}) + C$$

Dividing by  $e^{t/4}$ ,

$$v = 4\lambda(t - 4) + Ce^{-t/4}$$

The cage starts from rest, so  $v(0) = 0$ . Hence

$$0 = 4\lambda(0 - 4) + C$$

$$0 = -16\lambda + C$$

$$C = 16\lambda$$

Substituting back,

$$\begin{aligned} v &= 4\lambda(t - 4) + 16\lambda e^{-t/4} \\ &= 4\lambda(t - 4 + 4e^{-t/4}) \end{aligned}$$

So, for  $0 \leq t \leq 4$ ,

$$v = 4\lambda(t - 4 + 4e^{-t/4})$$

(ii) The velocity is continuous at  $t = 4$ , so the expressions from parts (c)(ii) and (e)(i) must be equal. From part (e)(i),

$$\begin{aligned} v(4) &= 4\lambda(4 - 4 + 4e^{-1}) \\ &= \frac{16\lambda}{e} \end{aligned}$$

From part (c)(ii),

$$v(4) = g(e - 1)$$

Therefore

$$\begin{aligned}\frac{16\lambda}{e} &= g(e - 1) \\ \lambda &= \frac{ge(e - 1)}{16} \\ &= \frac{g(e^2 - e)}{16}\end{aligned}$$

With  $g = 9.8$ ,

$$\lambda \approx \frac{9.8 \times e(e - 1)}{16} \approx 2.86$$

So

$$\lambda = \frac{ge(e - 1)}{16} \approx 2.86$$

(f) The total distance risen is

$$s = \int_0^4 v \, dt + \int_4^8 v \, dt$$

Using the expressions found for  $v$ ,

$$\begin{aligned}s_1 &= \int_0^4 4\lambda (t - 4 + 4e^{-t/4}) \, dt \\ &= 4\lambda \left( \int_0^4 t \, dt - \int_0^4 4 \, dt + \int_0^4 4e^{-t/4} \, dt \right) \\ &= 4\lambda \left( \left[ \frac{t^2}{2} \right]_0^4 - [4t]_0^4 + [-16e^{-t/4}]_0^4 \right) \\ &= 4\lambda \left( 8 - 16 + \left( -\frac{16}{e} + 16 \right) \right) \\ &= 4\lambda \left( 8 - \frac{16}{e} \right) \\ &= \left( 32 - \frac{64}{e} \right) \lambda\end{aligned}$$

For the second stage,

$$\begin{aligned}s_2 &= \int_4^8 g(e^{2-t/4} - 1) \, dt \\ &= g \left( \int_4^8 e^{2-t/4} \, dt - \int_4^8 1 \, dt \right) \\ &= g \left( [-4e^{2-t/4}]_4^8 - [t]_4^8 \right) \\ &= g((-4 + 4e) - 4) \\ &= 4g(e - 2)\end{aligned}$$

So

$$\begin{aligned}s &= s_1 + s_2 \\ &= \left( 32 - \frac{64}{e} \right) \lambda + 4g(e - 2)\end{aligned}$$

Substitute  $\lambda = \frac{ge(e-1)}{16}$ :

$$\begin{aligned}s &= \left(32 - \frac{64}{e}\right) \frac{ge(e-1)}{16} + 4g(e-2) \\&= \left(2 - \frac{4}{e}\right) ge(e-1) + 4g(e-2) \\&= (2e-4)g(e-1) + 4g(e-2) \\&= 2g(e-2)(e-1) + 4g(e-2) \\&= 2g(e-2)(e+1) \\&= 2g(e^2 - e - 2)\end{aligned}$$

Now use  $g = 9.8$ :

$$\begin{aligned}s &\approx 2(9.8)(e^2 - e - 2) \\&\approx 52.3 \text{ m}\end{aligned}$$

Therefore the total distance risen, to the nearest metre, is 52 m.

10. A short filling pulse in a buffer tank is to be modelled. The volume of liquid in the tank at time  $t$  seconds is  $V$  litres. The pulse starts when  $t = 0$ , and initially the tank is empty. After 1 second the volume is measured to be 4 litres. While the pulse is occurring,  $V$  is modelled by the differential equation

$$(3t - t^2) \frac{dV}{dt} = \frac{(3t - t^2)^2}{1 + (t - 1)^2} + (3 - 2t)V$$

- (a) By using an integrating factor show that, according to the model, while the pulse is occurring,

$$V = (3t - t^2) (\tan^{-1}(t - 1) + 2) \quad [6]$$

The pulse ends when the tank is again empty.

- (b) Determine the length of time that the pulse lasts according to the model [2]
- (c) Determine, according to the model, the rate of increase of the volume at the start of the pulse. Give your answer in an exact form [3]

### Solution

- (a) First rearrange the differential equation into linear form:

$$\frac{dV}{dt} - \frac{3 - 2t}{3t - t^2} V = \frac{3t - t^2}{1 + (t - 1)^2}$$

The integrating factor is

$$\mu(t) = e^{\int -\frac{3-2t}{3t-t^2} dt}$$

Since

$$\frac{d}{dt}(3t - t^2) = 3 - 2t$$

we have

$$\int -\frac{3 - 2t}{3t - t^2} dt = -\ln(3t - t^2)$$

Hence

$$\mu(t) = e^{-\ln(3t - t^2)} = \frac{1}{3t - t^2}$$

Multiplying the differential equation by this integrating factor gives

$$\frac{1}{3t - t^2} \frac{dV}{dt} - \frac{3 - 2t}{(3t - t^2)^2} V = \frac{1}{1 + (t - 1)^2}$$

The left-hand side is

$$\frac{d}{dt} \left( \frac{V}{3t - t^2} \right)$$

so

$$\frac{d}{dt} \left( \frac{V}{3t - t^2} \right) = \frac{1}{1 + (t - 1)^2}$$

Integrating with respect to  $t$ ,

$$\begin{aligned} \frac{V}{3t - t^2} &= \int \frac{1}{1 + (t - 1)^2} dt \\ &= \tan^{-1}(t - 1) + C \end{aligned}$$

Use the given condition  $V = 4$  when  $t = 1$ :

$$\begin{aligned}\frac{4}{3(1) - 1^2} &= \tan^{-1}(1 - 1) + C \\ \frac{4}{2} &= 0 + C \\ C &= 2\end{aligned}$$

Therefore

$$\frac{V}{3t - t^2} = \tan^{-1}(t - 1) + 2$$

and so

$$V = (3t - t^2) (\tan^{-1}(t - 1) + 2)$$

This is the required result.

(b) The pulse ends when the tank is empty again, so set  $V = 0$ :

$$(3t - t^2) (\tan^{-1}(t - 1) + 2) = 0$$

Now

$$-\frac{\pi}{2} < \tan^{-1}(t - 1) < \frac{\pi}{2}$$

so

$$2 - \frac{\pi}{2} < \tan^{-1}(t - 1) + 2 < 2 + \frac{\pi}{2}$$

Since  $2 - \frac{\pi}{2} > 0$ , the factor  $\tan^{-1}(t - 1) + 2$  is never zero. Hence

$$3t - t^2 = 0$$

so

$$\begin{aligned}t(3 - t) &= 0 \\ t &= 0 \text{ or } t = 3\end{aligned}$$

The pulse starts at  $t = 0$ , so it ends when  $t = 3$ . Therefore the pulse lasts 3 seconds.

(c) Differentiate the expression for  $V$ :

$$\begin{aligned}\frac{dV}{dt} &= \frac{d}{dt} [(3t - t^2) (\tan^{-1}(t - 1) + 2)] \\ &= (3 - 2t) (\tan^{-1}(t - 1) + 2) + (3t - t^2) \left( \frac{1}{1 + (t - 1)^2} \right)\end{aligned}$$

At the start of the pulse,  $t = 0$ :

$$\begin{aligned}\left. \frac{dV}{dt} \right|_{t=0} &= 3 (\tan^{-1}(-1) + 2) + 0 \\ &= 3 \left( -\frac{\pi}{4} + 2 \right) \\ &= 6 - \frac{3\pi}{4}\end{aligned}$$

Therefore the rate of increase of volume at the start is  $6 - \frac{3\pi}{4} \text{ L s}^{-1}$ .

11. A horticultural mixing tank initially contains 600 litres of pure water.

A fertiliser solution flows into the tank at a constant rate of 6 litres per minute. The incoming solution contains 2 grams of fertiliser in every litre of solution.

At the same time, liquid is pumped out of the well-stirred tank at a constant rate of 9 litres per minute.

It is assumed that the fertiliser instantly dissolves uniformly throughout the tank.

Given that there are  $x$  grams of fertiliser in the tank after  $t$  minutes,

- (a) Show that, while the tank still contains liquid, the situation can be modelled by the differential equation

$$\frac{dx}{dt} = 12 - \frac{3x}{200 - t} \quad [4]$$

- (b) Hence determine after how many minutes the concentration of fertiliser in the tank first reaches 1.5 grams per litre. [5]

- (c) Explain briefly one way in which the model could be refined. [1]

### Solution

- (a) Let  $V$  be the volume of liquid in the tank after  $t$  minutes.

Since liquid enters at 6 L/min and leaves at 9 L/min, the net change in volume is

$$\begin{aligned} V(t) &= 600 + (6 - 9)t \\ &= 600 - 3t \\ &= 3(200 - t) \end{aligned}$$

This is valid while the tank still contains liquid, so

$$600 - 3t > 0 \quad \Rightarrow \quad t < 200$$

The fertiliser enters at

$$6 \times 2 = 12 \text{ g/min}$$

The concentration in the tank at time  $t$  is

$$\frac{x}{V} = \frac{x}{600 - 3t} \text{ g/L}$$

So the fertiliser leaves at the rate

$$\begin{aligned} 9 \times \frac{x}{600 - 3t} &= \frac{9x}{600 - 3t} \\ &= \frac{9x}{3(200 - t)} \\ &= \frac{3x}{200 - t} \end{aligned}$$

Hence,

$$\begin{aligned} \frac{dx}{dt} &= \text{rate in} - \text{rate out} \\ &= 12 - \frac{3x}{200 - t} \end{aligned}$$

So the model is

$$\frac{dx}{dt} = 12 - \frac{3x}{200 - t}, \quad t < 200$$

(b) From part (a),

$$\frac{dx}{dt} + \frac{3}{200-t}x = 12$$

This is a linear differential equation. Its integrating factor is

$$\mu(t) = \exp\left(\int \frac{3}{200-t} dt\right) = \exp(-3\ln(200-t)) = (200-t)^{-3}$$

Multiplying the differential equation by  $(200-t)^{-3}$  gives

$$\begin{aligned}(200-t)^{-3} \frac{dx}{dt} + \frac{3x}{200-t} (200-t)^{-3} &= 12(200-t)^{-3} \\ \frac{d}{dt} (x(200-t)^{-3}) &= 12(200-t)^{-3}\end{aligned}$$

Integrating,

$$\begin{aligned}x(200-t)^{-3} &= \int 12(200-t)^{-3} dt \\ &= 6(200-t)^{-2} + C\end{aligned}$$

Therefore,

$$\begin{aligned}x &= (6(200-t)^{-2} + C)(200-t)^3 \\ &= 6(200-t) + C(200-t)^3\end{aligned}$$

Initially the tank contains pure water, so  $x = 0$  when  $t = 0$ . Substituting  $t = 0$  gives

$$\begin{aligned}0 &= 6(200) + C(200)^3 \\ 0 &= 1200 + 8,000,000C \\ C &= -\frac{1200}{8,000,000} \\ &= -\frac{3}{20000}\end{aligned}$$

Hence

$$x = 6(200-t) - \frac{3}{20000}(200-t)^3$$

The volume at time  $t$  is

$$V = 3(200-t)$$

So the concentration is

$$\begin{aligned}\frac{x}{V} &= \frac{6(200-t) - \frac{3}{20000}(200-t)^3}{3(200-t)} \\ &= 2 - \frac{(200-t)^2}{20000}\end{aligned}$$

We want this concentration to be 1.5 g/L, so

$$\begin{aligned}2 - \frac{(200-t)^2}{20000} &= 1.5 \\ \frac{(200-t)^2}{20000} &= 0.5 \\ (200-t)^2 &= 10000\end{aligned}$$

Since  $t < 200$ , we must have  $200-t > 0$ . Therefore

$$200-t = 100$$

So

$$t = 100$$

Therefore the concentration first reaches 1.5 g/L after 100 minutes.

- (c) One refinement would be to remove the assumption of perfect instantaneous mixing, and allow the fertiliser concentration to vary in different parts of the tank.

So a more realistic model could account for imperfect mixing.

- 12.** A mixing tank initially contains 2 litres of solution in which 4 grams of dye are dissolved.  
 A dye solution containing 10 grams of dye per litre is pumped into the tank at a constant rate of  $r$  litres per minute and mixes instantly and uniformly with the liquid already in the tank.  
 At the same time, liquid is removed from the tank at a constant rate of  $s$  litres per minute.  
 After  $t$  minutes, the tank contains  $m$  grams of dye.

- (a) (i) Show that the concentration of dye in the tank after  $t$  minutes is

$$\frac{m}{2 + (r - s)t}$$

grams per litre.

[2]

- (ii) Hence show that

$$\frac{dm}{dt} + \frac{sm}{2 + (r - s)t} = 10r$$

[2]

- (b) First, consider the case where  $s = r$ .

- (i) Solve the differential equation to find  $m$  in terms of  $r$  and  $t$ .

[4]

- (ii) Given that after 2 minutes the concentration of dye in the tank is 8 grams per litre, find  $r$ .

[2]

- (c) Now consider the case where  $s = 2r$ .

- (i) Explain why the differential equation in part (a)(ii) is no longer valid for  $t \geq \frac{2}{r}$ .

[1]

- (ii) Find the maximum mass of dye in the tank.

[9]

### Solution

- (a) (i) Initially the tank contains 2 litres.

In  $t$  minutes, liquid enters at  $r$  litres per minute, so volume added is  $rt$  litres.

In the same time, liquid leaves at  $s$  litres per minute, so volume removed is  $st$  litres.

Hence the volume in the tank after  $t$  minutes is

$$2 + rt - st = 2 + (r - s)t$$

Since the mass of dye is  $m$  grams, the concentration is

$$\frac{m}{2 + (r - s)t}$$

So the concentration of dye after  $t$  minutes is  $\frac{m}{2 + (r - s)t}$  grams per litre.

- (ii) The rate at which dye enters the tank is

$$10 \times r = 10r$$

grams per minute.

From part (i), the concentration in the tank is  $\frac{m}{2 + (r - s)t}$  grams per litre, so the rate at which dye leaves is

$$s \cdot \frac{m}{2 + (r - s)t}$$

grams per minute.

Therefore

$$\frac{dm}{dt} = \text{rate in} - \text{rate out}$$

so

$$\frac{dm}{dt} = 10r - \frac{sm}{2 + (r - s)t}$$

Rearranging gives

$$\frac{dm}{dt} + \frac{sm}{2 + (r - s)t} = 10r$$

(b) (i) If  $s = r$ , then part (a)(ii) becomes

$$\frac{dm}{dt} + \frac{rm}{2 + (r - r)t} = 10r$$

so

$$\frac{dm}{dt} + \frac{r}{2}m = 10r$$

Also, initially there are 4 grams of dye, so

$$m(0) = 4$$

This is a linear differential equation. Its integrating factor is

$$\exp\left(\int \frac{r}{2} dt\right) = e^{rt/2}$$

Multiplying the differential equation by  $e^{rt/2}$  gives

$$e^{rt/2} \frac{dm}{dt} + \frac{r}{2} e^{rt/2} m = 10r e^{rt/2}$$

The left-hand side is the derivative of  $me^{rt/2}$ , so

$$\frac{d}{dt} \left( me^{rt/2} \right) = 10r e^{rt/2}$$

Integrating,

$$\begin{aligned} me^{rt/2} &= \int 10r e^{rt/2} dt \\ &= 10r \cdot \frac{2}{r} e^{rt/2} + C \\ &= 20e^{rt/2} + C \end{aligned}$$

Therefore

$$m = 20 + Ce^{-rt/2}$$

Using  $m(0) = 4$ ,

$$\begin{aligned} 4 &= 20 + Ce^0 \\ 4 &= 20 + C \\ C &= -16 \end{aligned}$$

Hence

$$m = 20 - 16e^{-rt/2}$$

(ii) When  $s = r$ , the volume is constant and equal to

$$2 + (r - r)t = 2$$

So the concentration is

$$\frac{m}{2} = \frac{20 - 16e^{-rt/2}}{2} = 10 - 8e^{-rt/2}$$

After 2 minutes, the concentration is 8 grams per litre, so

$$8 = 10 - 8e^{-r}$$

Solving,

$$\begin{aligned}8e^{-r} &= 2 \\e^{-r} &= \frac{1}{4} \\-r &= \ln\left(\frac{1}{4}\right) \\r &= \ln 4\end{aligned}$$

Therefore

$$r = \ln 4 \approx 1.39$$

So the required inflow rate is  $r = \ln 4$  litres per minute.

- (c) (i) If  $s = 2r$ , then the volume after  $t$  minutes is

$$2 + (r - 2r)t = 2 - rt$$

At

$$t = \frac{2}{r}$$

this volume is 0, so the tank is empty.

For  $t \geq \frac{2}{r}$ , the denominator in the differential equation is 0 or negative, so the model is no longer physically valid.

Hence the differential equation is not valid for  $t \geq \frac{2}{r}$  because the tank is empty at  $t = \frac{2}{r}$ .

- (ii) With  $s = 2r$ , the differential equation from part (a)(ii) is

$$\frac{dm}{dt} + \frac{2rm}{2 + (r - 2r)t} = 10r$$

so

$$\frac{dm}{dt} + \frac{2r}{2 - rt}m = 10r$$

with

$$m(0) = 4$$

We solve this for  $0 \leq t < \frac{2}{r}$ .

The integrating factor is

$$\begin{aligned}\exp\left(\int \frac{2r}{2 - rt} dt\right) &= \exp(-2 \ln(2 - rt)) \\&= (2 - rt)^{-2}\end{aligned}$$

Multiplying through by  $(2 - rt)^{-2}$  gives

$$(2 - rt)^{-2} \frac{dm}{dt} + \frac{2r}{2 - rt} (2 - rt)^{-2} m = 10r(2 - rt)^{-2}$$

The left-hand side is

$$\frac{d}{dt} (m(2 - rt)^{-2})$$

So

$$\frac{d}{dt} (m(2 - rt)^{-2}) = 10r(2 - rt)^{-2}$$

Integrating,

$$\begin{aligned}m(2 - rt)^{-2} &= \int 10r(2 - rt)^{-2} dt \\&= \frac{10}{2 - rt} + C\end{aligned}$$

Therefore

$$m = 10(2 - rt) + C(2 - rt)^2$$

Using  $m(0) = 4$ ,

$$4 = 10(2) + C(2)^2$$

$$4 = 20 + 4C$$

$$4C = -16$$

$$C = -4$$

Hence

$$m = 10(2 - rt) - 4(2 - rt)^2$$

Expanding,

$$\begin{aligned} m &= 20 - 10rt - 4(4 - 4rt + r^2t^2) \\ &= 20 - 10rt - 16 + 16rt - 4r^2t^2 \\ &= 4 + 6rt - 4r^2t^2 \end{aligned}$$

This is a quadratic in  $t$  with negative coefficient of  $t^2$ , so it has a maximum at its stationary point.

Differentiate:

$$\frac{dm}{dt} = 6r - 8r^2t$$

Set this equal to 0:

$$6r - 8r^2t = 0$$

$$6r = 8r^2t$$

$$6 = 8rt$$

$$t = \frac{3}{4r}$$

Since  $\frac{3}{4r} < \frac{2}{r}$ , this occurs before the tank becomes empty, so it is valid.

Now substitute  $t = \frac{3}{4r}$  into  $m = 4 + 6rt - 4r^2t^2$ :

$$\begin{aligned} m_{\max} &= 4 + 6r \left( \frac{3}{4r} \right) - 4r^2 \left( \frac{3}{4r} \right)^2 \\ &= 4 + \frac{18}{4} - 4r^2 \cdot \frac{9}{16r^2} \\ &= 4 + \frac{18}{4} - \frac{36}{16} \\ &= 4 + \frac{9}{2} - \frac{9}{4} \\ &= \frac{16}{4} + \frac{18}{4} - \frac{9}{4} \\ &= \frac{25}{4} \end{aligned}$$

Therefore the maximum mass of dye in the tank is

$$\frac{25}{4} \text{ g}$$

- 13.** The concentration  $C$ , in suitable units, of a drug in a patient's bloodstream at time  $t$  hours after an intravenous infusion begins is to be modelled. Initially, the concentration is zero, and its long-term value is  $M$ .

(a) One simple model assumes that the rate of change of concentration is directly proportional to  $M - C$ .

(i) Formulate a differential equation for this model. [1]

(ii) Verify that  $C = M(1 - e^{-kt})$ , where  $k$  is a positive constant, satisfies this differential equation, the initial condition and the long-term condition. [3]

An alternative model uses, for  $t > 0$ , the differential equation

$$\frac{dC}{dt} - \frac{t+2}{t(1+t+t^2)} C = S(t),$$

where  $S(t)$  is a function of  $t$ .

(b) Find the integrating factor for this differential equation, showing that it can be written in the form

$$\frac{1+t+t^2}{t^2} \quad [8]$$

(c) Suppose that  $S(t) = 0$ .

(i) Show that

$$C = \frac{Mt^2}{1+t+t^2} \quad [4]$$

(ii) Find the time predicted by this model for the concentration to reach half its long-term value. Give your answer correct to the nearest minute. [2]

(d) Now suppose that

$$S(t) = \frac{3t^2 e^{-t}}{1+t+t^2}$$

Show that

$$C = \frac{Mt^2 - 3t^2 e^{-t}}{1+t+t^2} \quad [5]$$

(e) It is found that the long-term concentration is 10, and  $C$  reaches half this value after 105 minutes. Determine which of the models proposed in parts (c) and (d) is more consistent with this data. [2]

### Solution

(a) (i) Since the rate of change is directly proportional to  $M - C$ ,

$$\frac{dC}{dt} = k(M - C), \quad k > 0$$

(ii) Take

$$C = M(1 - e^{-kt})$$

Differentiate:

$$\begin{aligned}\frac{dC}{dt} &= \frac{d}{dt} [M(1 - e^{-kt})] \\ &= M(0 - (-k)e^{-kt}) \\ &= Mke^{-kt}\end{aligned}$$

Also,

$$\begin{aligned}M - C &= M - M(1 - e^{-kt}) \\ &= M - M + Me^{-kt} \\ &= Me^{-kt}\end{aligned}$$

Hence,

$$\frac{dC}{dt} = Mke^{-kt} = k(Me^{-kt}) = k(M - C)$$

so the differential equation is satisfied.

For the initial condition,

$$C(0) = M(1 - e^0) = M(1 - 1) = 0$$

For the long-term condition, since  $k > 0$ ,  $e^{-kt} \rightarrow 0$  as  $t \rightarrow \infty$ . Therefore

$$\lim_{t \rightarrow \infty} C = \lim_{t \rightarrow \infty} M(1 - e^{-kt}) = M(1 - 0) = M$$

So  $C = M(1 - e^{-kt})$  satisfies the differential equation, the initial condition, and the long-term condition.

**(b)** Write the equation in the linear form

$$\frac{dC}{dt} + P(t)C = S(t)$$

where

$$P(t) = -\frac{t+2}{t(1+t+t^2)}$$

To find the integrating factor, first decompose  $P(t)$  into partial fractions:

$$-\frac{t+2}{t(1+t+t^2)} = \frac{A}{t} + \frac{Bt+D}{1+t+t^2}$$

Multiplying through by  $t(1+t+t^2)$  gives

$$\begin{aligned}-(t+2) &= A(1+t+t^2) + t(Bt+D) \\ &= A + At + At^2 + Bt^2 + Dt \\ &= A + (A+D)t + (A+B)t^2\end{aligned}$$

Comparing coefficients with  $-2 - t + 0t^2$ :

$$\begin{aligned}A &= -2 \\ A + D &= -1 \\ A + B &= 0\end{aligned}$$

So

$$A = -2, \quad D = 1, \quad B = 2$$

Therefore

$$P(t) = -\frac{2}{t} + \frac{2t+1}{1+t+t^2}$$

The integrating factor is

$$\mu(t) = e^{\int P(t) dt}$$

Since  $t > 0$ ,

$$\begin{aligned}\int P(t) \, dt &= \int \left( -\frac{2}{t} + \frac{2t+1}{1+t+t^2} \right) \, dt \\ &= -2 \ln t + \ln(1+t+t^2)\end{aligned}$$

Hence

$$\begin{aligned}\mu(t) &= e^{-2 \ln t + \ln(1+t+t^2)} \\ &= e^{\ln(1+t+t^2)} e^{-2 \ln t} \\ &= (1+t+t^2)t^{-2} \\ &= \frac{1+t+t^2}{t^2}\end{aligned}$$

So the integrating factor is  $\frac{1+t+t^2}{t^2}$ .

(c) (i) When  $S(t) = 0$ , the differential equation becomes

$$\frac{dC}{dt} - \frac{t+2}{t(1+t+t^2)}C = 0$$

Using the integrating factor

$$\mu(t) = \frac{1+t+t^2}{t^2}$$

gives

$$\frac{d}{dt}(\mu C) = 0$$

So

$$\mu C = A$$

where  $A$  is a constant. Thus

$$\begin{aligned}C &= \frac{A}{\mu} \\ &= \frac{A}{(1+t+t^2)/t^2} \\ &= \frac{At^2}{1+t+t^2}\end{aligned}$$

Now use the long-term value:

$$\begin{aligned}\lim_{t \rightarrow \infty} C &= \lim_{t \rightarrow \infty} \frac{At^2}{1+t+t^2} \\ &= \lim_{t \rightarrow \infty} \frac{A}{t^{-2} + t^{-1} + 1} \\ &= A\end{aligned}$$

Since the long-term concentration is  $M$ , we must have  $A = M$ .

Therefore

$$C = \frac{Mt^2}{1+t+t^2}$$

(ii) Half the long-term value is  $\frac{M}{2}$ . Set

$$\frac{Mt^2}{1+t+t^2} = \frac{M}{2}$$

Cancelling  $M$  and solving:

$$\begin{aligned}\frac{t^2}{1+t+t^2} &= \frac{1}{2} \\ 2t^2 &= 1+t+t^2 \\ t^2 - t - 1 &= 0\end{aligned}$$

Hence

$$t = \frac{1 \pm \sqrt{5}}{2}$$

Since  $t > 0$ ,

$$t = \frac{1 + \sqrt{5}}{2} \text{ hours} \approx 1.618 \text{ hours}$$

Converting to minutes:

$$1.618 \times 60 \approx 97.1$$

So the concentration reaches half its long-term value after 97 minutes, to the nearest minute.

(d) Using the integrating factor from part (b),

$$\mu(t) = \frac{1 + t + t^2}{t^2}$$

and

$$S(t) = \frac{3t^2 e^{-t}}{1 + t + t^2}$$

Then

$$\begin{aligned} \mu(t)S(t) &= \frac{1 + t + t^2}{t^2} \cdot \frac{3t^2 e^{-t}}{1 + t + t^2} \\ &= 3e^{-t} \end{aligned}$$

So the differential equation becomes

$$\frac{d}{dt}(\mu C) = 3e^{-t}$$

Integrating:

$$\begin{aligned} \mu C &= \int 3e^{-t} dt \\ &= -3e^{-t} + A \end{aligned}$$

Hence

$$\begin{aligned} C &= \frac{-3e^{-t} + A}{\mu} \\ &= \frac{t^2(-3e^{-t} + A)}{1 + t + t^2} \\ &= \frac{At^2 - 3t^2 e^{-t}}{1 + t + t^2} \end{aligned}$$

Now use the long-term value. As  $t \rightarrow \infty$ ,

$$\frac{At^2}{1 + t + t^2} \rightarrow A$$

and, using the fact that  $\lim_{t \rightarrow \infty} t^2 e^{-t} = 0$ ,

$$\frac{3t^2 e^{-t}}{1 + t + t^2} \rightarrow 0$$

Therefore

$$\lim_{t \rightarrow \infty} C = A$$

Since the long-term concentration is  $M$ ,  $A = M$ .

So

$$C = \frac{Mt^2 - 3t^2 e^{-t}}{1 + t + t^2}$$

as required.

- (e) Here  $M = 10$ , so half the long-term concentration is 5, reached after 105 minutes = 1.75 hours.  
From part (c), model (c) predicts the half-value time to be 97 minutes.  
For model (d), at  $t = 1.75$ ,

$$\begin{aligned} C &= \frac{10t^2 - 3t^2e^{-t}}{1 + t + t^2} \\ &= \frac{10(1.75)^2 - 3(1.75)^2e^{-1.75}}{1 + 1.75 + (1.75)^2} \\ &\approx \frac{30.625 - 1.597}{5.8125} \\ &\approx 4.99 \end{aligned}$$

This is essentially 5.

Therefore model (d) is more consistent with the data.

14. A drinks manufacturer uses a mixing tank that initially contains 4 litres of water. A flavouring liquid, which is 60% fruit concentrate by volume, is pumped into the tank at a constant rate of  $p$  litres per minute. At the same time, the contents of the tank are thoroughly mixed and drawn off at a constant rate of  $q$  litres per minute.

After  $t$  minutes, let  $x$  litres be the amount of fruit concentrate in the tank. You may assume that mixing is instantaneous and uniform.

- (a) (i) Show that the proportion of fruit concentrate in the tank after  $t$  minutes is

$$\frac{x}{4 + (p - q)t} \quad [2]$$

- (ii) Hence show that

$$\frac{dx}{dt} + \frac{qx}{4 + (p - q)t} = \frac{3p}{5} \quad [2]$$

- (b) Now consider the case where  $q = p$ .

- (i) Solve the differential equation to find  $x$  in terms of  $p$  and  $t$ . [4]

- (ii) Given that after 4 minutes the liquid in the tank is 30% fruit concentrate, find the value of  $p$ . [2]

- (c) Now consider the case where  $q = 2p$ .

- (i) Explain why the differential equation in part (a)(ii) is no longer valid for  $t \geq \frac{4}{p}$ . [1]

- (ii) Find the maximum amount of fruit concentrate in the tank before the tank empties. [9]

### Solution

- (a) (i) After  $t$  minutes, the volume in the tank is

$$4 + pt - qt = 4 + (p - q)t$$

Since there are  $x$  litres of fruit concentrate in the tank, the proportion of fruit concentrate is

$$\frac{x}{4 + (p - q)t}$$

- (ii) The flavouring liquid entering the tank is 60% concentrate, so the rate at which concentrate enters is

$$0.6p = \frac{3p}{5}$$

From part (i), the proportion of concentrate in the tank is  $\frac{x}{4 + (p - q)t}$ , so the rate at which concentrate leaves is

$$q \left( \frac{x}{4 + (p - q)t} \right) = \frac{qx}{4 + (p - q)t}$$

Therefore

$$\begin{aligned} \frac{dx}{dt} &= \text{rate in} - \text{rate out} \\ &= \frac{3p}{5} - \frac{qx}{4 + (p - q)t} \end{aligned}$$

Rearranging,

$$\frac{dx}{dt} + \frac{qx}{4 + (p - q)t} = \frac{3p}{5}$$

Hence the differential equation is shown.

(b) (i) If  $q = p$ , then the differential equation from part (a)(ii) becomes

$$\frac{dx}{dt} + \frac{px}{4 + (p - p)t} = \frac{3p}{5}$$

so

$$\frac{dx}{dt} + \frac{p}{4}x = \frac{3p}{5}$$

Initially the tank contains only water, so

$$x(0) = 0$$

The integrating factor is

$$\mu(t) = e^{\int \frac{p}{4} dt} = e^{pt/4}$$

Multiplying the differential equation by  $e^{pt/4}$  gives

$$\begin{aligned} e^{pt/4} \frac{dx}{dt} + \frac{p}{4} e^{pt/4} x &= \frac{3p}{5} e^{pt/4} \\ \frac{d}{dt} (x e^{pt/4}) &= \frac{3p}{5} e^{pt/4} \end{aligned}$$

Integrating,

$$\begin{aligned} x e^{pt/4} &= \int \frac{3p}{5} e^{pt/4} dt \\ &= \frac{3p}{5} \cdot \frac{4}{p} e^{pt/4} + C \\ &= \frac{12}{5} e^{pt/4} + C \end{aligned}$$

Hence

$$x = \frac{12}{5} + C e^{-pt/4}$$

Using  $x(0) = 0$ ,

$$0 = \frac{12}{5} + C$$

so

$$C = -\frac{12}{5}$$

Therefore

$$x = \frac{12}{5} (1 - e^{-pt/4})$$

(ii) When  $q = p$ , the total volume stays constant at 4 litres.

After 4 minutes, the liquid is 30% concentrate, so the amount of concentrate is

$$x = 0.3 \times 4 = \frac{6}{5}$$

Substitute  $t = 4$  into the formula from part (i):

$$\frac{12}{5} (1 - e^{-p}) = \frac{6}{5}$$

Multiply through by 5:

$$12(1 - e^{-p}) = 6$$

Hence

$$\begin{aligned}1 - e^{-p} &= \frac{1}{2} \\ e^{-p} &= \frac{1}{2}\end{aligned}$$

Taking natural logarithms,

$$-p = \ln\left(\frac{1}{2}\right) = -\ln 2$$

so

$$p = \ln 2$$

Therefore  $p = \ln 2 \approx 0.693$  litres  $\text{min}^{-1}$ .

- (c) (i) If  $q = 2p$ , then the volume after  $t$  minutes is

$$4 + (p - 2p)t = 4 - pt$$

At

$$t = \frac{4}{p}$$

this volume is 0, so the tank is empty. For  $t \geq \frac{4}{p}$ , the denominator  $4 - pt$  is zero or negative, which is not physically meaningful.

Therefore the differential equation is only valid for  $0 \leq t < \frac{4}{p}$ .

- (ii) With  $q = 2p$ , the differential equation becomes

$$\frac{dx}{dt} + \frac{2px}{4 - pt} = \frac{3p}{5}$$

for  $0 \leq t < \frac{4}{p}$ , with

$$x(0) = 0$$

The integrating factor is

$$\mu(t) = e^{\int \frac{2p}{4-pt} dt}$$

Now let  $u = 4 - pt$ , so  $du = -p dt$ . Then

$$\begin{aligned}\int \frac{2p}{4 - pt} dt &= -2 \int \frac{1}{u} du \\ &= -2 \ln u \\ &= -2 \ln(4 - pt)\end{aligned}$$

Hence

$$\mu(t) = e^{-2 \ln(4-pt)} = (4 - pt)^{-2}$$

Multiply the differential equation by  $(4 - pt)^{-2}$ :

$$(4 - pt)^{-2} \frac{dx}{dt} + \frac{2p}{4 - pt} (4 - pt)^{-2} x = \frac{3p}{5} (4 - pt)^{-2}$$

The left-hand side is

$$\frac{d}{dt} (x(4 - pt)^{-2})$$

so

$$\frac{d}{dt} (x(4 - pt)^{-2}) = \frac{3p}{5} (4 - pt)^{-2}$$

Integrating,

$$x(4 - pt)^{-2} = \int \frac{3p}{5} (4 - pt)^{-2} dt$$

Again let  $u = 4 - pt$ , so  $du = -p dt$ . Then

$$\begin{aligned}\int \frac{3p}{5}(4 - pt)^{-2} dt &= -\frac{3}{5} \int u^{-2} du \\ &= -\frac{3}{5} \left( \frac{u^{-1}}{-1} \right) + C \\ &= \frac{3}{5} u^{-1} + C \\ &= \frac{3}{5(4 - pt)} + C\end{aligned}$$

Therefore

$$x(4 - pt)^{-2} = \frac{3}{5(4 - pt)} + C$$

Multiply by  $(4 - pt)^2$ :

$$x = \frac{3}{5}(4 - pt) + C(4 - pt)^2$$

Use  $x(0) = 0$ :

$$\begin{aligned}0 &= \frac{3}{5}(4) + C(4)^2 \\ 0 &= \frac{12}{5} + 16C\end{aligned}$$

so

$$C = -\frac{3}{20}$$

Hence

$$\begin{aligned}x &= \frac{3}{5}(4 - pt) - \frac{3}{20}(4 - pt)^2 \\ &= \frac{12}{5} - \frac{3pt}{5} - \frac{3}{20}(16 - 8pt + p^2t^2) \\ &= \frac{12}{5} - \frac{3pt}{5} - \frac{12}{5} + \frac{6pt}{5} - \frac{3p^2t^2}{20} \\ &= \frac{3pt}{5} - \frac{3p^2t^2}{20}\end{aligned}$$

So

$$x = \frac{3pt}{5} - \frac{3p^2t^2}{20} = \frac{3}{20}pt(4 - pt)$$

To find the maximum value, differentiate:

$$\frac{dx}{dt} = \frac{3p}{5} - \frac{3p^2t}{10}$$

Set this equal to 0:

$$\begin{aligned}\frac{3p}{5} - \frac{3p^2t}{10} &= 0 \\ \frac{3p}{10}(2 - pt) &= 0\end{aligned}$$

Since  $p > 0$ , this gives

$$2 - pt = 0$$

so

$$t = \frac{2}{p}$$

This lies in the interval  $0 \leq t < \frac{4}{p}$ , and since the quadratic has negative coefficient of  $t^2$ , this gives the maximum.

Now substitute  $t = \frac{2}{p}$ :

$$\begin{aligned}x_{\max} &= \frac{3p}{5} \left( \frac{2}{p} \right) - \frac{3p^2}{20} \left( \frac{2}{p} \right)^2 \\&= \frac{6}{5} - \frac{3}{20} \cdot 4 \\&= \frac{6}{5} - \frac{12}{20} \\&= \frac{6}{5} - \frac{3}{5} \\&= \frac{3}{5}\end{aligned}$$

Therefore the maximum amount of fruit concentrate is  $\frac{3}{5}$  litres, occurring at  $t = \frac{2}{p}$  minutes.